Chapter 4
Signal Processing

Minghui Shi
Xiamen University

Processing Stages --- Review

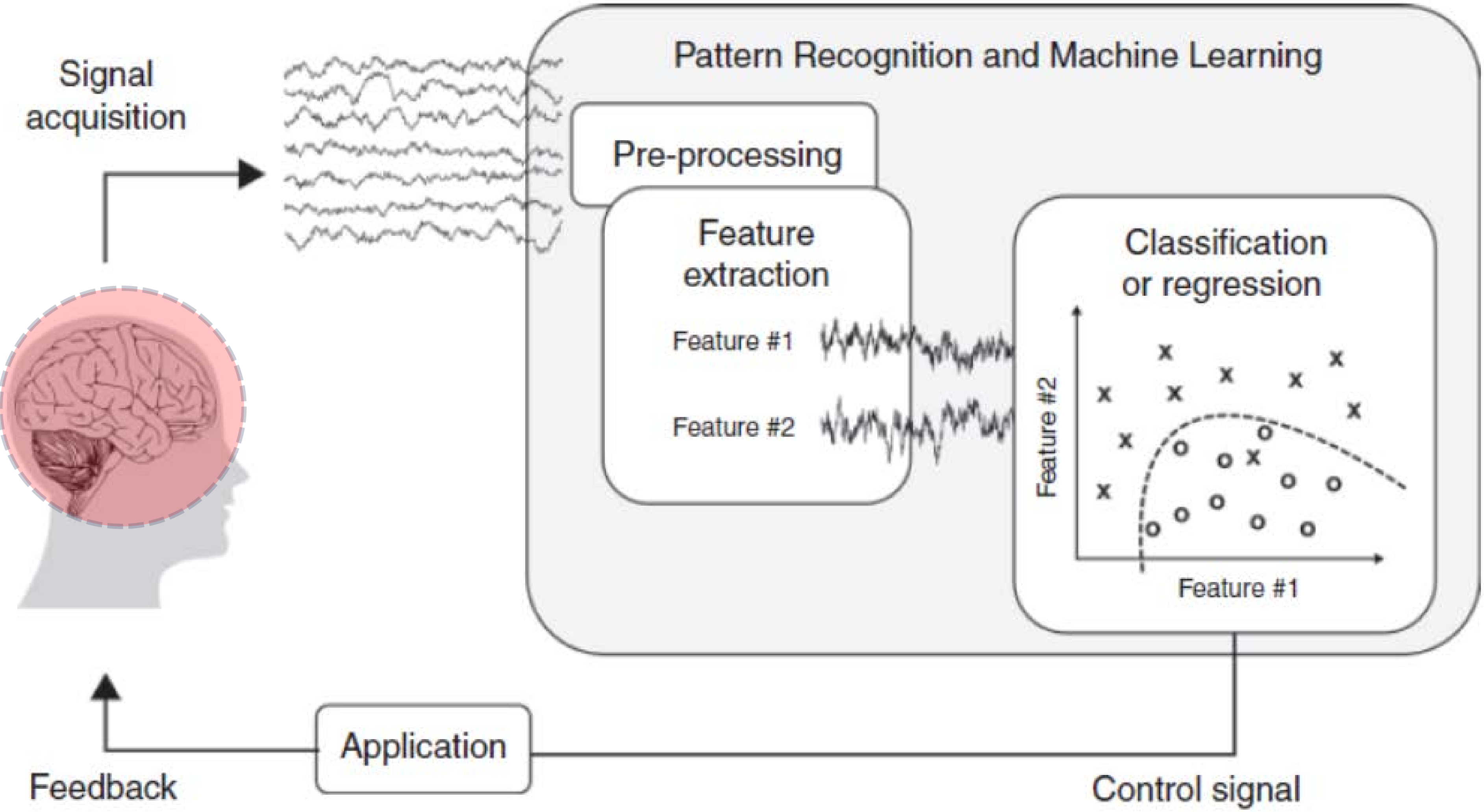


Figure 1.1. **Basic components of a brain-computer interface (BCI).**

Processing Stages --- Review

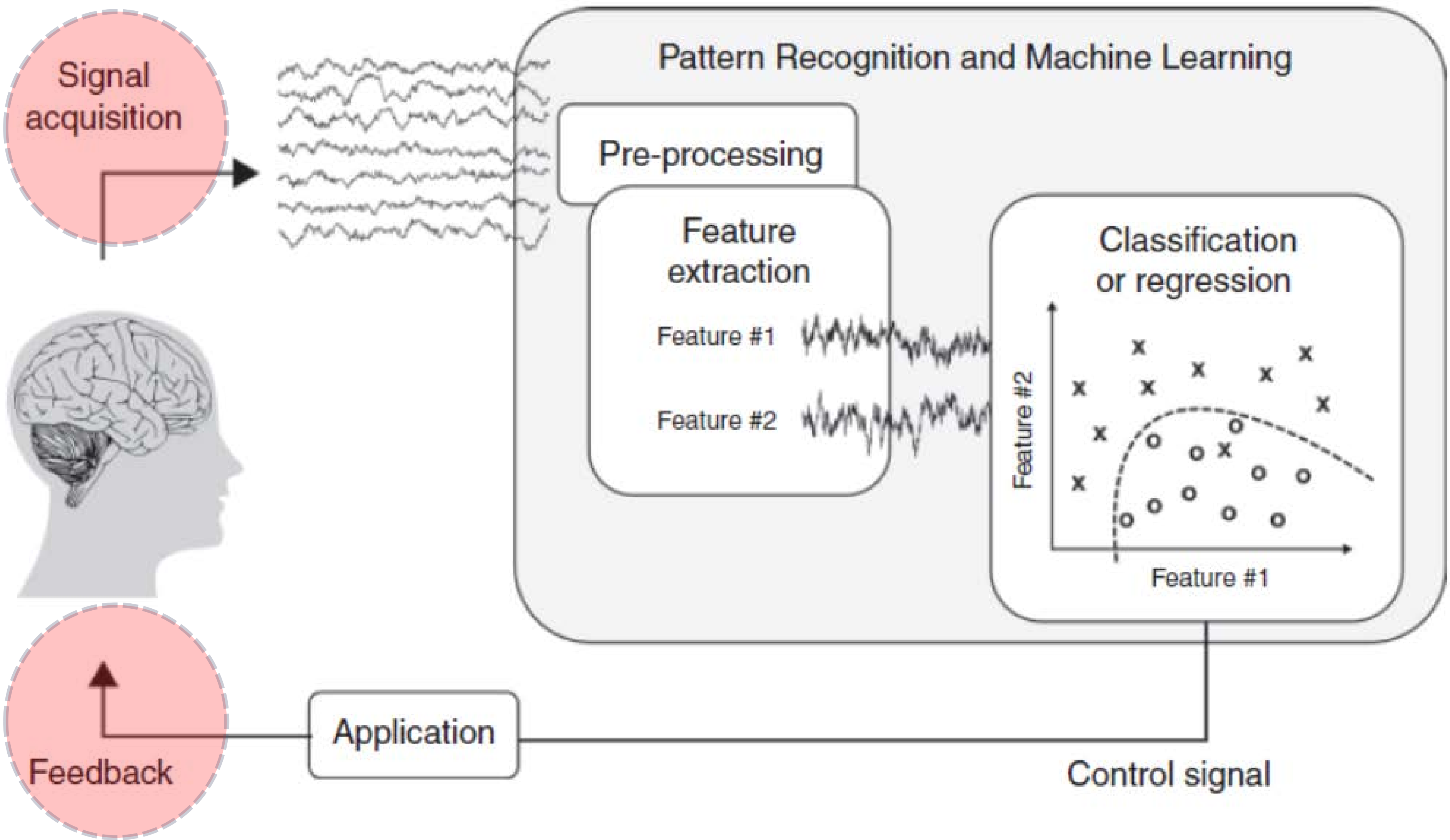


Figure 1.1. **Basic components of a brain-computer interface (BCI).**

Processing Stages --- Review

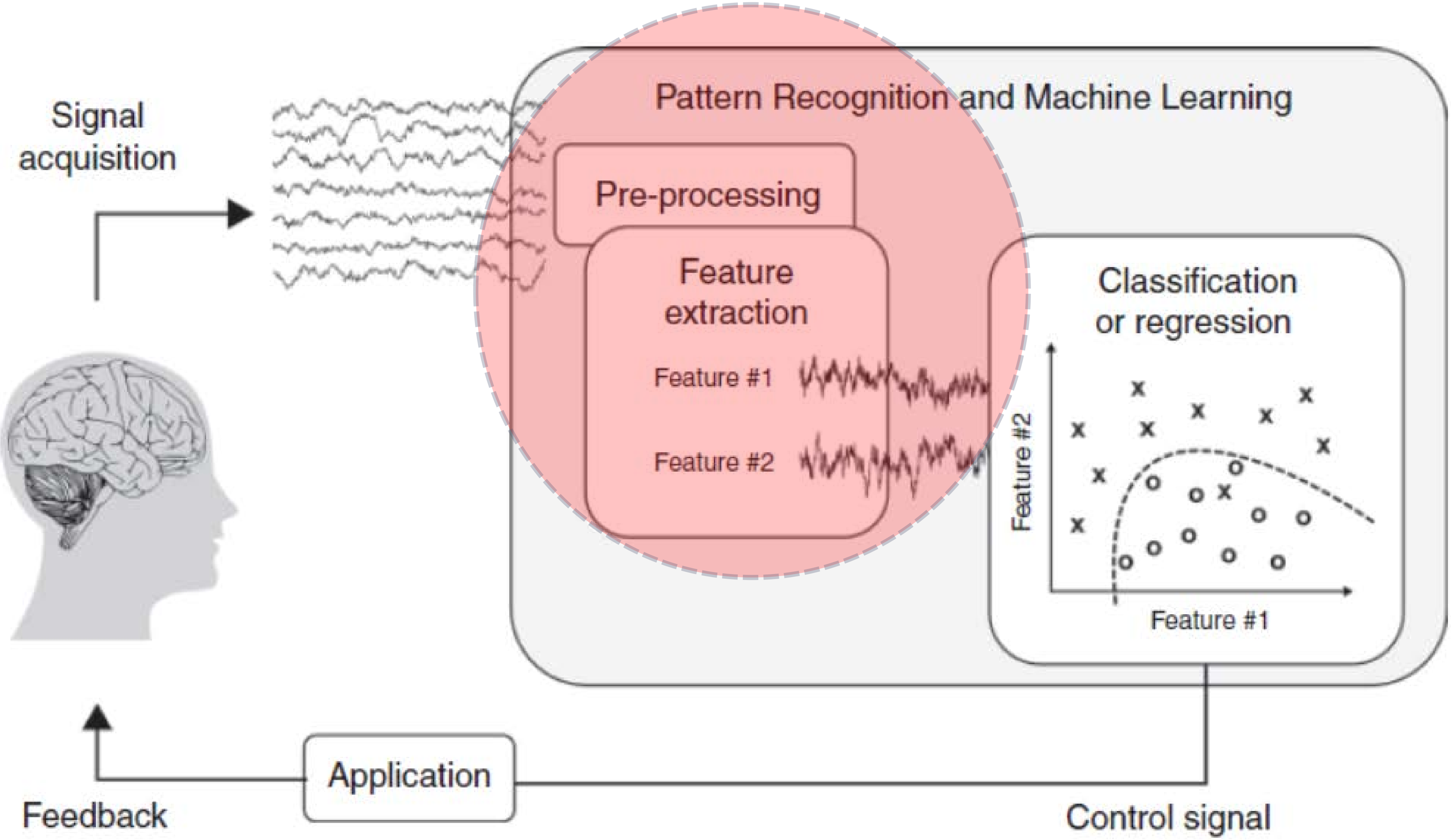


Figure 1.1. **Basic components of a brain-computer interface (BCI).**

Content & Topics

- 1. Introduction
- Part I Background
  - 2. Basic Neuroscience
  - 3. Recording and Stimulating the Brain
  - 4. Signal Processing
  - 5. Machine Learning
- Part II Putting It All Together
- Part III Major Types of BCIs
- Part IV Applications and Ethics

Goal

- In this chapter, we review the **signal-processing methods** applied to recorded brain signals in BCIs **for tasks** ranging from **extracting spikes** from the raw signals recorded from invasive electrodes to **extracting features** for classification.

- For many of the techniques, we use EEG as the noninvasive recording modality to illustrate the concepts involved, although the techniques could be applied to signals from other sources as well such as ECoG and MEG.

Content

Content

Content

Chapter 4
Signal Processing

4.1 Spike Sorting

Lecturer: Minghui Shi, Xiamen University

4.1 Spike Sorting

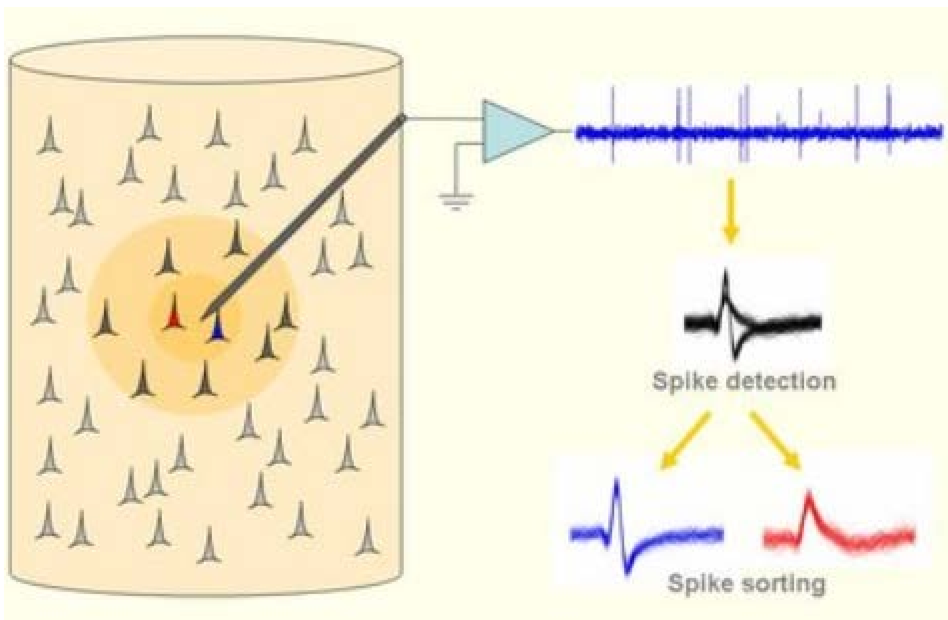
- What is spike sorting?
- What is multiunit hash or neural hash?
- What is the simplest method?
- What is the better method?
- What is the recent trend?

4.1 Spike Sorting

- Video

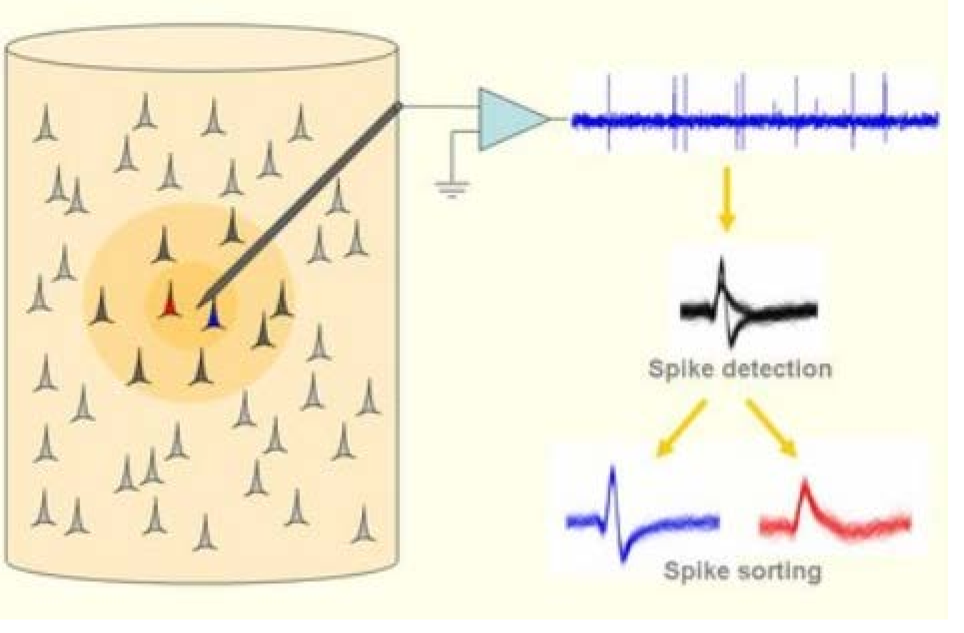
4.1 Spike Sorting
What is spike sorting?

- Isolate and extract the spikes being emitted by a single neuron per recording electrode.
- Extract spikes from a particular single neuron from hash.



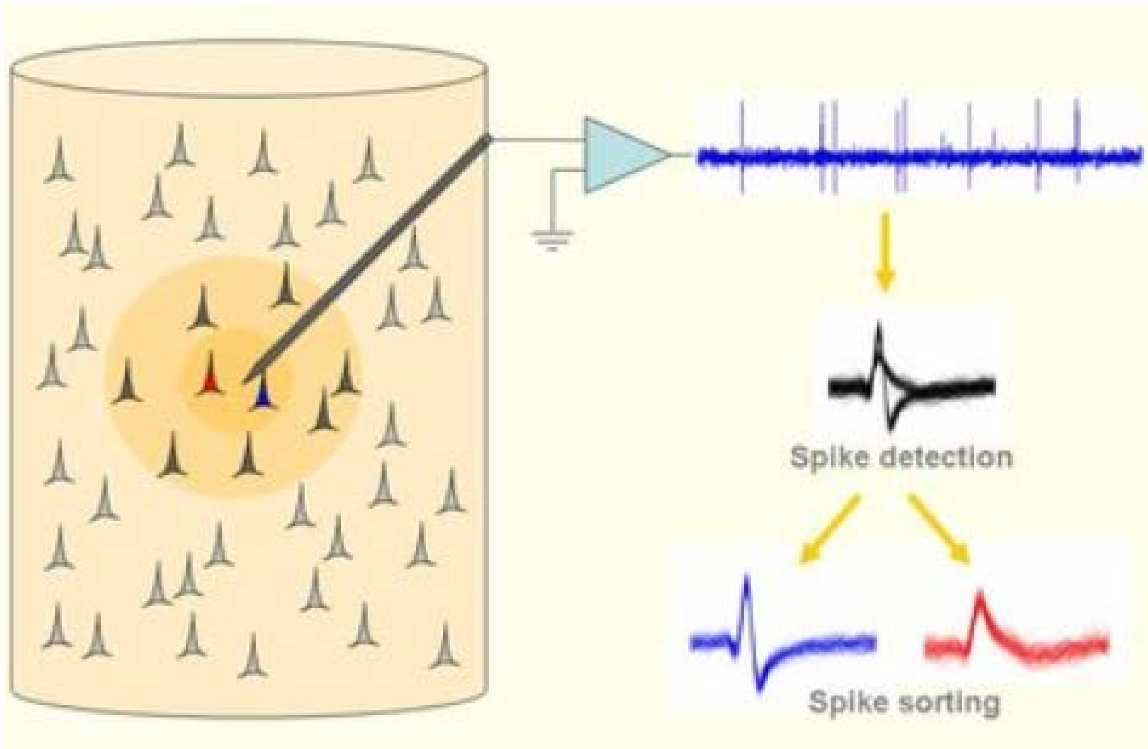
4.1 Spike Sorting
What is multiunit hash or neural hash?

- A mixture of signals from several neighboring neurons, with spikes from closer neurons producing larger amplitude deflections in the recorded signal.



4.1 Spike Sorting
What is the simplest method?

- Classify spikes according to their peak amplitude.



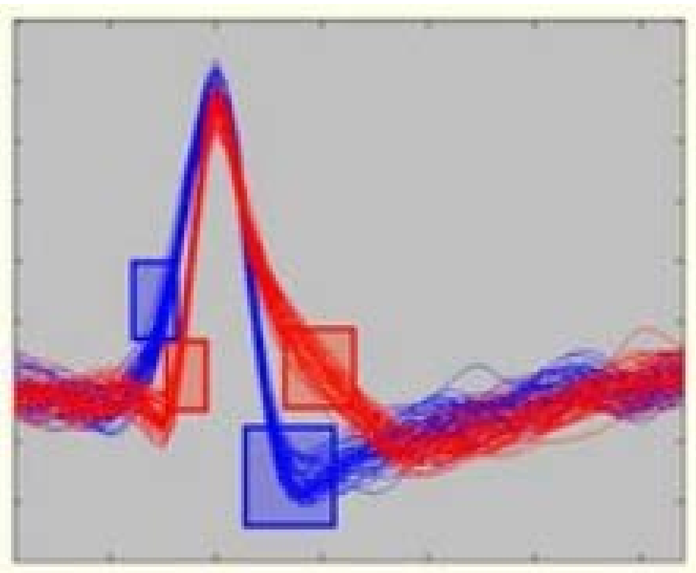
4.1 Spike Sorting
What is the better method?

- **Window Discriminator Method**
  1. visually examines the data;
  2. places windows on aligned recordings of spikes of the same shape;
  3. assigns all future spikes crossing one or more of these windows to the same neuron.

Detail can be seen at
http://www.scholarpedia.org/article/Spike_sorting

4.1 Spike Sorting
What is the better method? (Con.)

Detail can be seen at
http://www.scholarpedia.org/article/Spike_sorting



Window discriminators.
Spike shapes are plotted superimposed and they are clustered using time-amplitude windows manually set by the user. In this case, the two red windows determine the cluster of spikes in red and the blue windows determine the cluster of spikes in blue.

4.1 Spike Sorting
What is the recent trend?

- **Drawback** of Window Discriminator Method:
  - Manually labelling

- **Recent trend:**
  - Automatically clustering

Chapter 4
Signal Processing

4.2 Frequency Domain Analysis

Lecturer: Minghui Shi, Xiamen University

4.2 Frequency Domain Analysis

- What makes the analysis in the frequency domain particular useful?

- A: **amplitude/power changes in the oscillatory activity** caused by large populations of neurons.

4.2 Frequency Domain Analysis

4.2.1 Fourier Analysis

4.2.1 Fourier Analysis

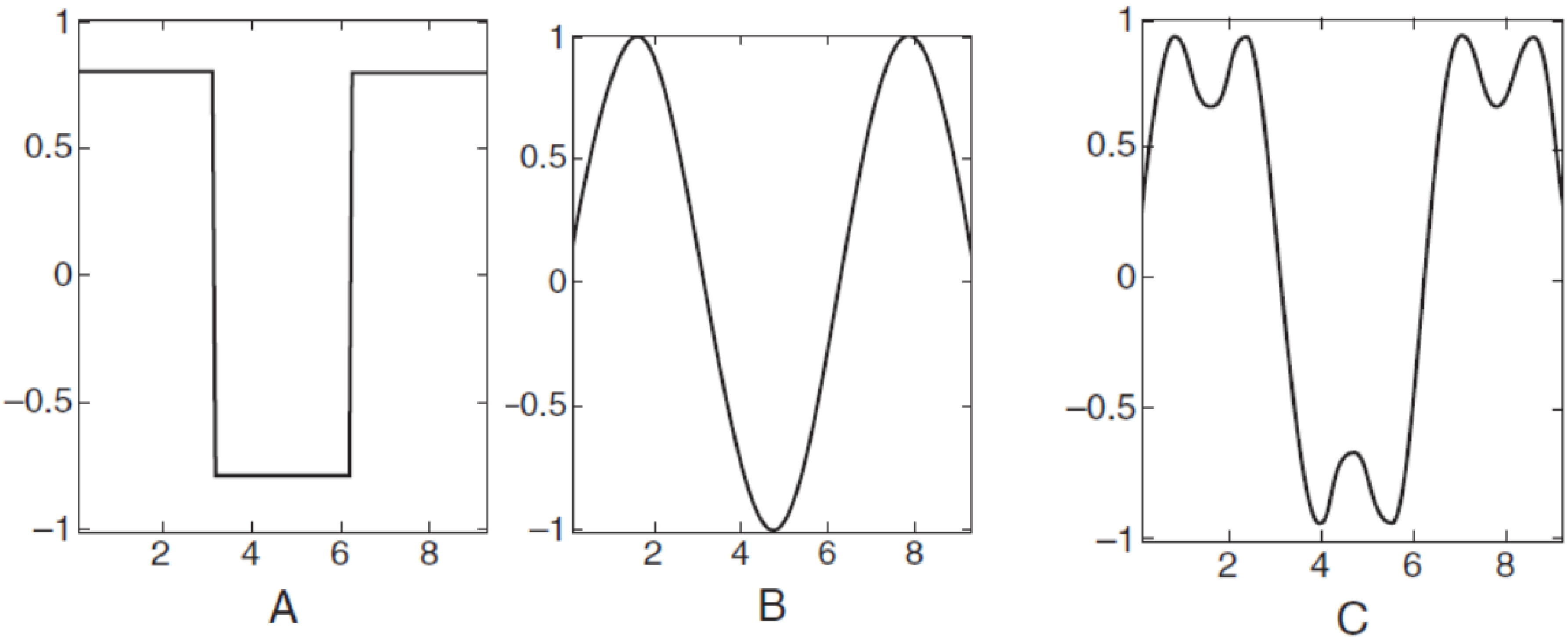
- Basic Idea
- Example

4.2.1 Fourier Analysis
Basic Idea

- **Basic Idea:** decompose a signal into a weighted sum of sine and cosine waves of different frequencies.
- In specific, Fourier analysis involves decomposing a time-varying signal s(t), defined over an interval t = -T/2 to t = T/2, into a weighted sum of sine and cosine waves of different frequencies

$$
\begin{aligned}
s(t) &= \frac{a_0}{2} + a_1 \cos(\omega t) + a_2 \cos(2\omega t) + \ldots + b_1 \sin(\omega t) + b_2 \sin(2\omega t) + \ldots \\
&= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t) \\
&= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(2\pi nft) + \sum_{n=1}^{\infty} b_n \sin(2\pi nft)
\end{aligned}
$$

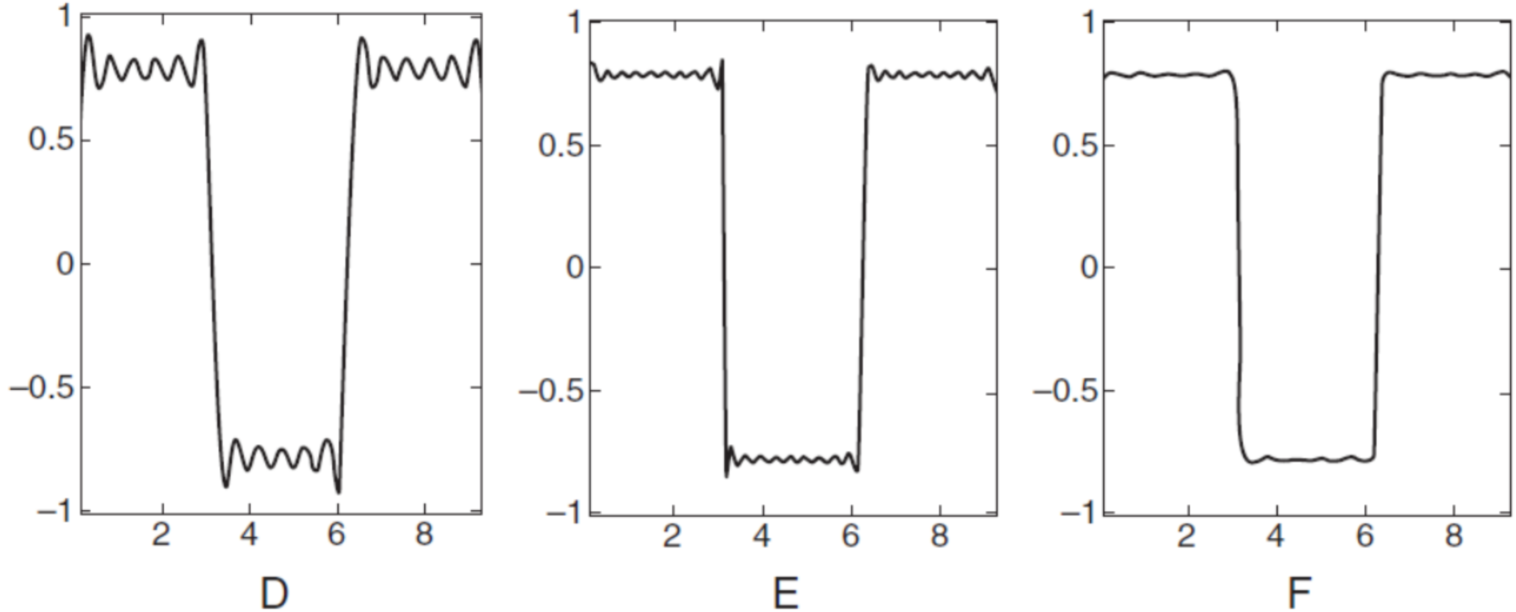
4.2.1 Fourier Analysis
Example



$\sin(x)$.

$\sin(x) + (1/3)\sin(3x)$.

http://open.163.com/newview/movie/free?pid=MD5B0JHKB&mid=MD5B69QMV



sin(x) + (1/3)sin(3x) +…+ (1/51)*sin(51x)

$$s(t)=\frac{a_0}{2}+a_1\cos(\omega t)+a_2\cos(2\omega t)+\ldots+b_1\sin(\omega t)+b_2\sin(2\omega t)+\ldots$$

$$=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos(n\omega t)+\sum_{n=1}^{\infty}b_n\sin(n\omega t)$$

$$=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos(2\pi nft)+\sum_{n=1}^{\infty}b_n\sin(2\pi nft)$$

4.2 Frequency Domain Analysis

4.2.2 Discrete Fourier Transform (DFT)

Lecturer: Minghui Shi, Xiamen University

4.2.2 Discrete Fourier Transform(DFT)

- For BCI applications, the brain signals are typically sampled at discrete time intervals. The Fourier series discussed above can be modified to apply to a discretely sampled signal as well.

- The Fourier series discussed above can be modified as **discrete Fourier transform (DFT)** to apply to a discretely sampled signal as well.

4.2 Frequency Domain Analysis

4.2.3 Fast Fourier Transform (FFT)

4.2.3 Fast Fourier Transform (FFT)

- One can compute the DFT based on its definition , but for a signal with T points, this takes approximately T^2 arithmetical operations. The running time of the algorithm is thus quadratic in signal size T. For very large T (e.g., in the millions), this can be quite slow.

- The fast Fourier transform (FFT) is a more efficient way of computing the DFT. It runs in time approximately T log T, which can result in huge savings in computation time for large sizes T.
- Many signal-processing packages, such as MATLAB, provide the function tool for FFT.

4.2 Frequency Domain Analysis

4.2.4 Spectral Features

Lecturer: Minghui Shi, Xiamen University

4.2.4 Spectral Features

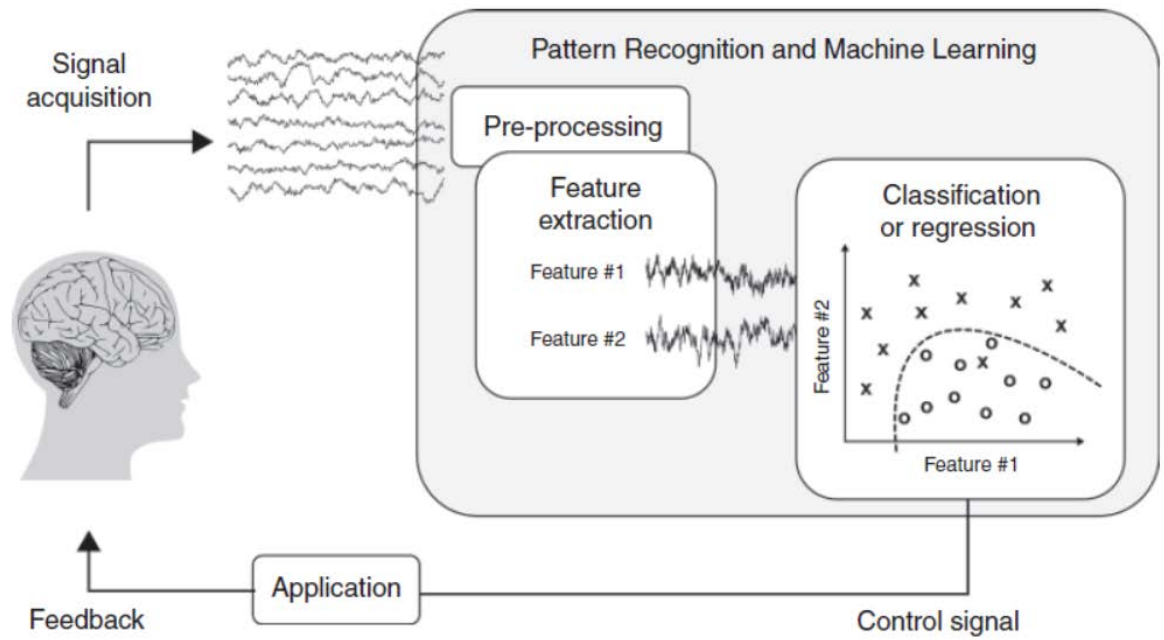
- Many BCI systems rely on **features extracted from the power spectrum** of a brain signal such as EEG or ECoG over a time interval.

- The **power spectrum** is first computed using an FFT, and the power in a particular frequency band is used as a spectral feature in further analysis (such as classification).

4.2.4 Spectral Features (Con.) Application Example1

- **Use the power in the mu band as a feature** in a BCI to allow a subject to move a cursor using motor imagery.

- Video



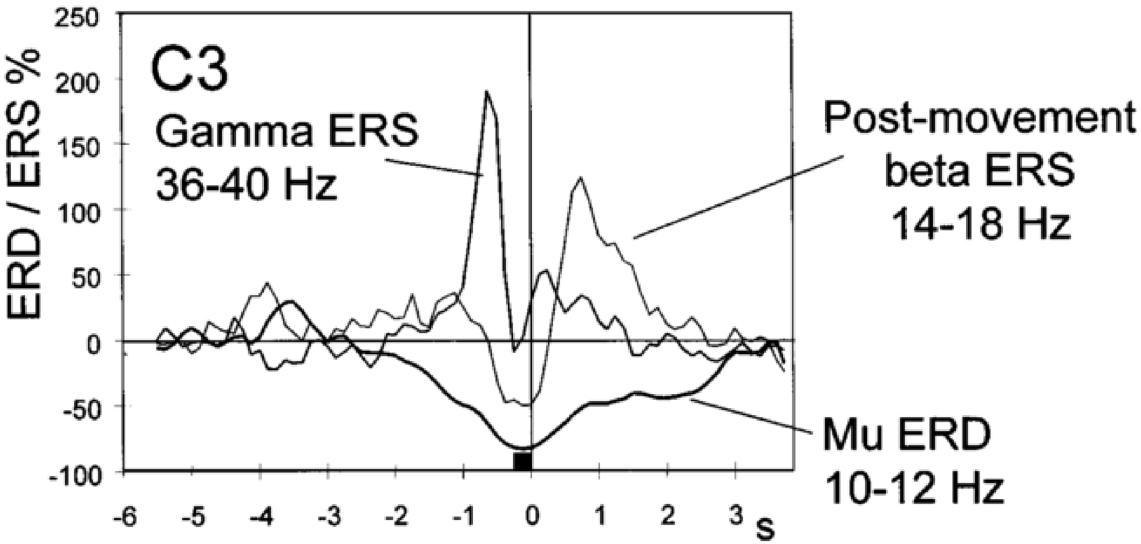
4.2.4 Spectral Features (Con.) Application Example1(Con.)

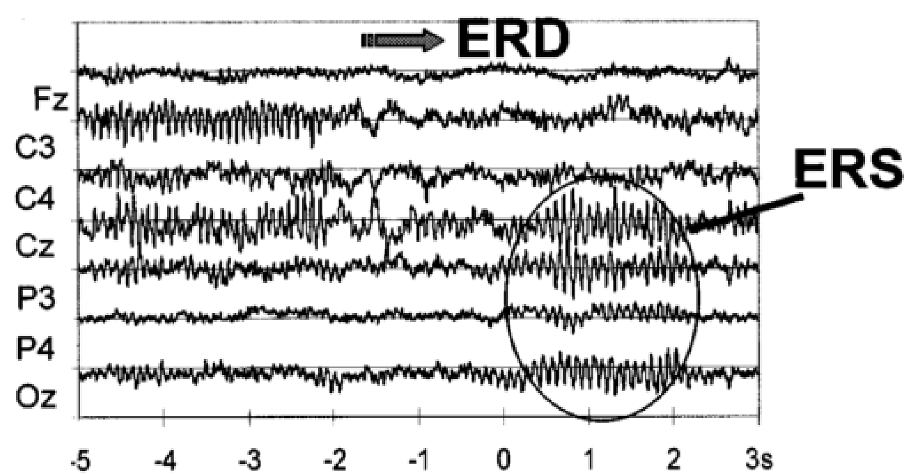
- **Use the power in the mu band as a feature** in a BCI to allow a subject to move a cursor using motor imagery.

- Video

- Paper:
  1. G. Pfurtscheller and C. Neuper, "**Motor imagery and direct brain-computer communication**," *Proceedings of the IEEE*, vol. 89, pp. 1123-1134, 2001.
  2. G. Pfurtscheller, C. Brunner, A. Schlögl, and F. L. Da Silva, "**Mu rhythm (de) synchronization and EEG single-trial classification of different motor imagery tasks**," *Neuroimage*, vol. 31, pp. 153-159, 2006.

4.2.4 Spectral Features (Con.) Application Example1(Con.)

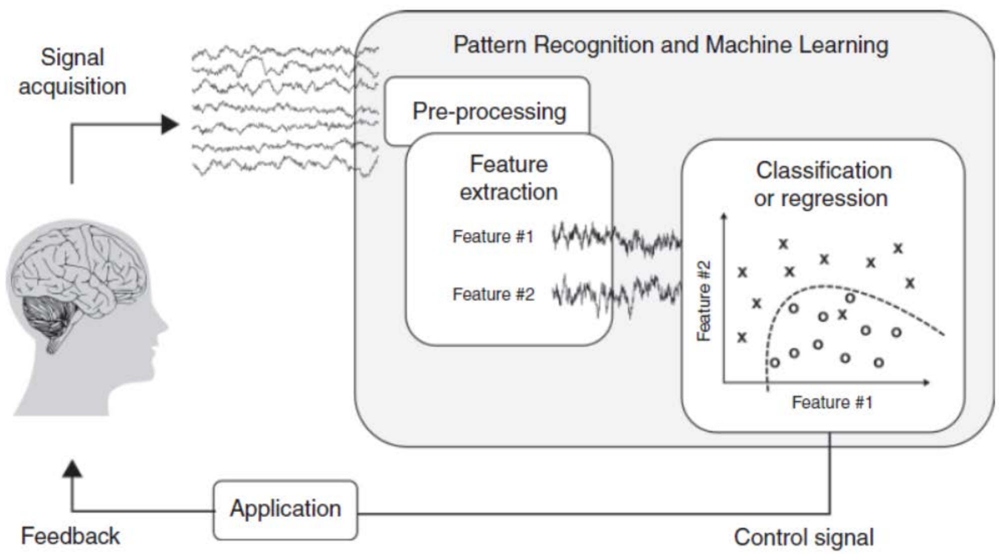




1. G. Pfurtscheller and C. Neuper, "Motor imagery and direct brain-computer communication," *Proceedings of the IEEE*, vol. 89, pp. 1123-1134, 2001.

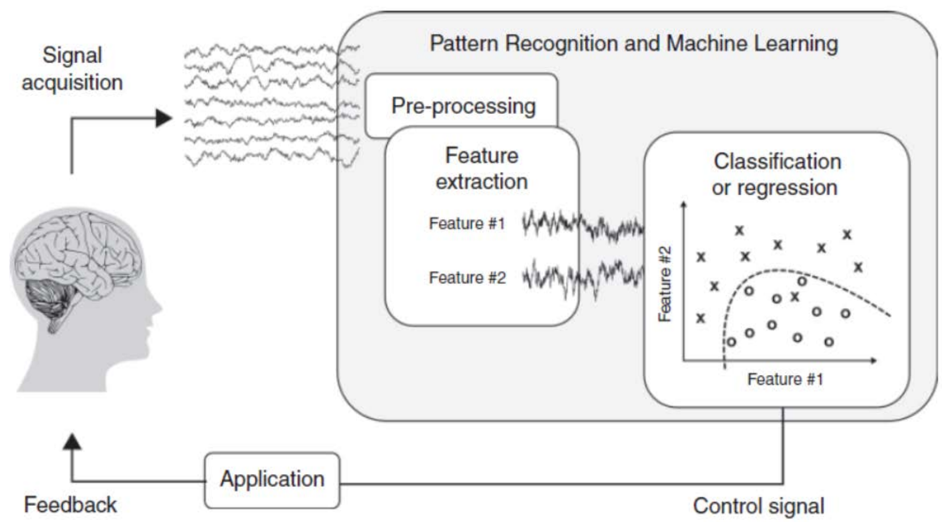
4.2.4 Spectral Features (Con.) Application Example 2

- Use motor screening to **find subject-specific frequency bands**:
  1. the subject is asked to perform a variety of movements;
  2. the frequency bands that exhibit robust changes in power during movement are then utilized in subsequent BCI experiments involving movement imagery.



4.2.4 Spectral Features (Con.) Application Example 3

- Utilize a bank of spectral features and allow a machine learning algorithm to automatically select features that enhance classification accuracy on test data.



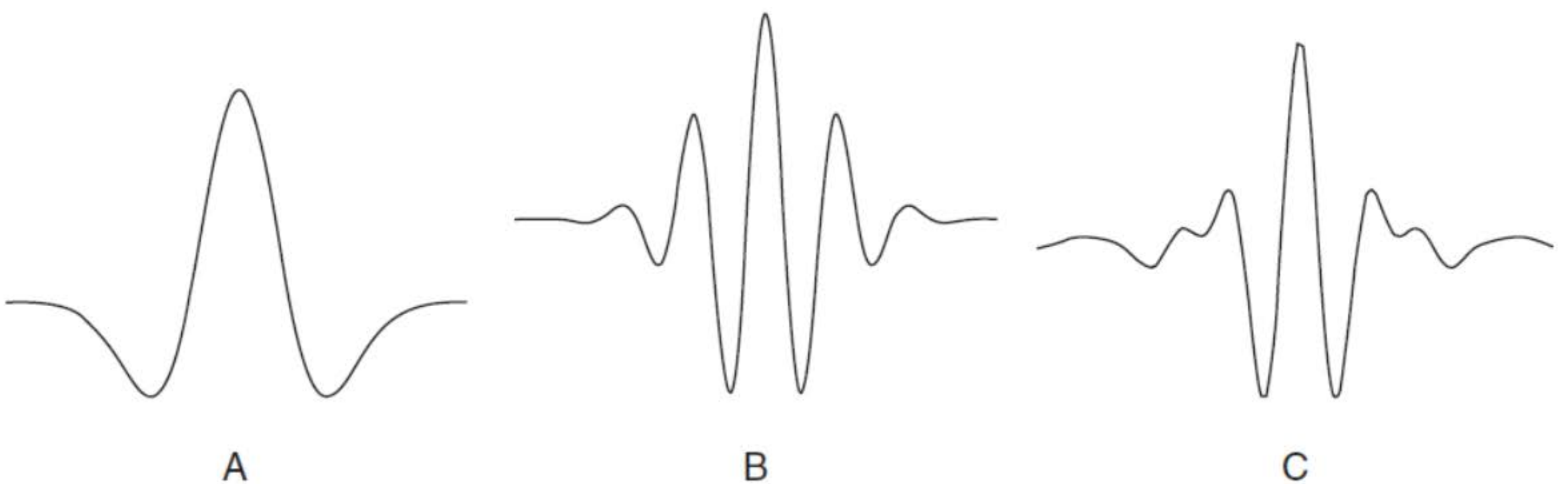
Chapter 4
Signal Processing

4.3 Wavelet Analysis

Lecturer: Minghui Shi, Xiamen University

4.3 Wavelet Analysis

- As in the case of the Fourier transform, the wavelet transform represents the original signal as a linear combination of basis functions, in this case, the wavelets (see Figure 4.5).



4.3 Wavelet Analysis (Con.)

- Analysis of the signal is performed using the corresponding **wavelet coefficients**.

- Most current **signal-processing packages** include the wavelet transform as one of the available options and provide a variety of choices for the mother wavelet.

Chapter 4
Signal Processing

4.4 Time Domain Analysis

Lecturer: Minghui Shi, Xiamen University

4.4 Time Domain Analysis
Content

- 4.4.1 Hjorth Parameters
- 4.4.2 Fractal Dimension
- 4.4.3 Autoregressive (AR) Modeling
- 4.4.4 Bayesian Filtering
- 4.4.5 Kalman Filtering
- 4.4.6 Particle Filtering

4.4 Time Domain Analysis

4.4.1 Hjorth Parameters

Lecturer: Minghui Shi, Xiamen University

4.4.1 Hjorth Parameters

- (*read this section with the following questions.*)
- What's the Hjorth Parameters?
- How to calculate them?

4.4 Time Domain Analysis

4.4.2 Fractal Dimension

Lecturer: Minghui Shi, Xiamen University

4.4.2 Fractal Dimension

- (*read this section with the following questions.*)
- What is the fractal signal?
- What is the Fractal Dimension?
- How to compute the Fractal Dimension using Higuchi's method? (*optional material*)

4.4 Time Domain Analysis

4.4.3 Autoregressive (AR) Modeling

Lecturer: Minghui Shi, Xiamen University

4.4.3 Autoregressive (AR) Modeling

- (*read this section with the following questions.*)
- What's the fact on which AR rely?
- What's the formula of the AR model?
- What's the AAR? What's the difference between AAR and AR?
- Why do we use AAR rather than AR in BCI?
- Can the coefficients of ARR act as features?

4.4 Time Domain Analysis

4.4.4 Bayesian Filtering

Lecturer: Minghui Shi, Xiamen University

4.4.4 Bayesian Filtering

- (*read this section with the following questions.*)
- Why do we use Bayesian Filtering?
- What's the Bayes' Theorem or Bayes' Rule?
- What's the significance of Bayes' Theorem or Bayes' Rule?
- What's the relationship between Bayesian Filtering and Kalman Filtering as well as Particle Filtering?
- What's the relationship between Bayesian Filtering and Deep Bayesian Network (DBN)?
- What's the DBN?

4.4 Time Domain Analysis

4.4.5 Kalman Filtering

Lecturer: Minghui Shi, Xiamen University

4.4.5 Kalman Filtering

- (*read this section with the following questions.*)
- What's the relationship between Bayesian Filtering and Kalman Filtering?
- What's the goal of Kalman Filtering?
- How to deal with Kalman Filtering?
- What's Kalman Gain?
- How to understand the formula of Kaman Filtering and Kalman Gain?

4.4 Time Domain Analysis

4.4.6 Particle Filtering

Lecturer: Minghui Shi, Xiamen University

4.4.6 Particle Filtering

- (*read this section with the following questions.*)
- What's the relationship between Bayesian Filtering and Particle Filtering?
- What's the difference between Kalman Filtering and Particle Filtering?
- What does the "particle" means?

Chapter 4
Signal Processing

4.5 Spatial Filtering

Lecturer: Minghui Shi, Xiamen University

4.5 Spatial Filtering

- **What is Spatial Filtering?**

- **What is the goal of Spatial Filtering?**

4.5 Spatial Filtering

- **What is Spatial Filtering?**
  - Spatial filtering techniques take as input brain signals recorded from several different locations (or “channels”) and transform them in one of several ways.

4.5 Spatial Filtering

- **What is the goal of Spatial Filtering?**

1. Enhancing local activity
2. Reducing noise that is common across channels
3. Decreasing the dimensionality of the data
4. Identifying hidden sources
5. Finding projections that maximize discrimination between different classes

4.5 Spatial Filtering Content

4.5 Spatial Filtering

4.5.1 Bipolar, Laplacian, and Common Average Referencing

Lecturer: Minghui Shi, Xiamen University

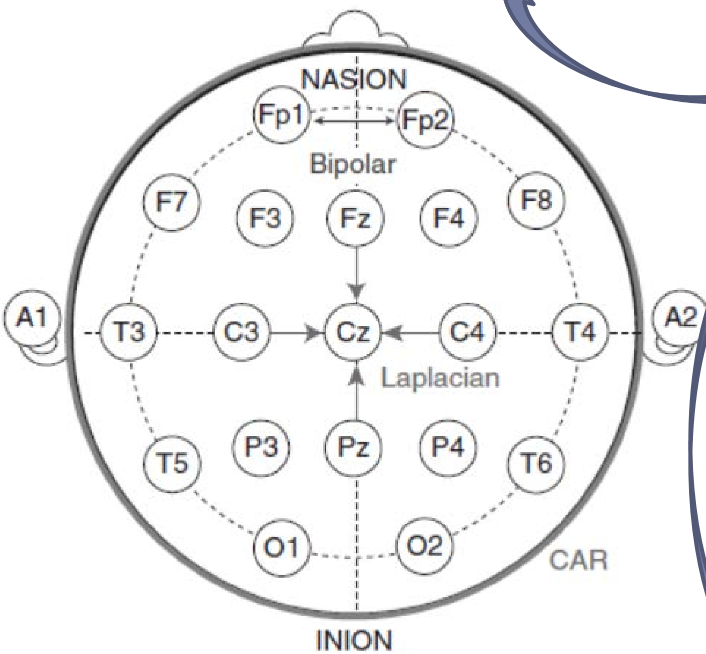
4.5.1 Bipolar, Laplacian, and Common Average Referencing

- (*read this section with the following questions.*)
- What's the definition and role of Bipolar Signal?
- What's the definition and role of Laplacian Filtering?
- What's the definition and role of Common Average Referencing?

4.5.1 Bipolar, Laplacian, and Common Average Referencing

Highlight differences between two electrodes.

bipolar signals $\tilde{\mathbf{s}}_{i,j} = \mathbf{s}_i - \mathbf{s}_j$



Laplacian filtering

$$\tilde{\mathbf{s}}_i = \mathbf{s}_i - \frac{1}{4}\sum_{i\in\Theta}\mathbf{s}_i$$

Common Average Referencing

$$\tilde{\mathbf{s}}_i = \mathbf{s}_i - \frac{1}{N}\sum_{i=1}^{N}\mathbf{s}_i$$

Extract local activity at electrode i, substract common activity.

4.5 Spatial Filtering

4.5.2 Principal Component Analysis (PCA)

Lecturer: Minghui Shi, Xiamen University

4.5.2 Principal Component Analysis (PCA)

- (*read this section with the following questions*.)
- What is the data set dealt by PCA like?
- What's the goal of PCA?
- How to achieve the goal?
- Why are EEG signals amenable to dimensionality reduction?
- What does "principal component" mean?

4.5.2 Principal Component Analysis (PCA)
Goal of PCA

Multi-channel Signals $\xrightarrow{\text{Spatial Filtering}}$ Transformed Signals

Multi-channel Signals (L-dimensional) (Redundant) $\xrightarrow{\text{PCA}}$ Multi-channel Signals (M-dimensional, M<<L) (Decorrelated)

4.5.2 Principal Component Analysis (PCA)

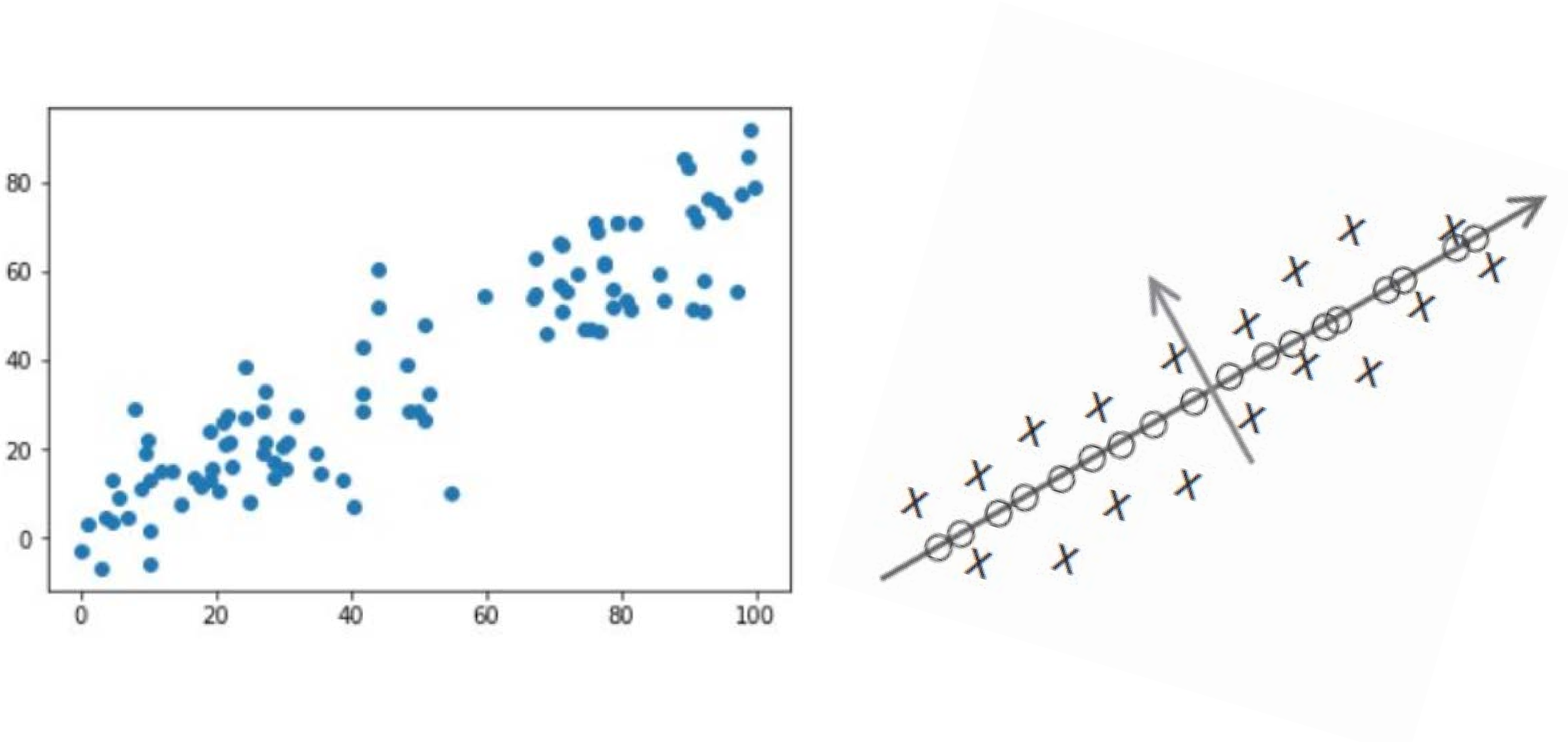


Figure 4.9. **Principal component analysis (PCA).**

4.5.2 Principal Component Analysis (PCA)

- Most natural signals, including brain signals recorded from multiple locations, tend to be redundant and are therefore amenable to dimensionality reduction.

- For example, in the case of EEG measurements from N electrodes placed on the head, measurements from nearby electrodes may be correlated or there may be underlying rhythms that appear across multiple electrodes.

4.5.2 Principal Component Analysis (PCA)

- Such redundancies can be exploited by PCA, which attempts to find the dominant directions of variability in the data.

- Once these dominant directions corresponding to a low-dimensional "subspace" of the original L-dimensional space have been found, new data points can be projected along these "principal" directions.
  - Each projection is called a "principal component,"

4.5.2 Principal Component Analysis (PCA)

- In summary, PCA produces a vector **a** that is both low-dimensional and decorrelated.

- Such a representation can be a useful "feature vector" for classification and other types of analysis in BCI applications.

4.5.2 Principal Component Analysis (PCA)

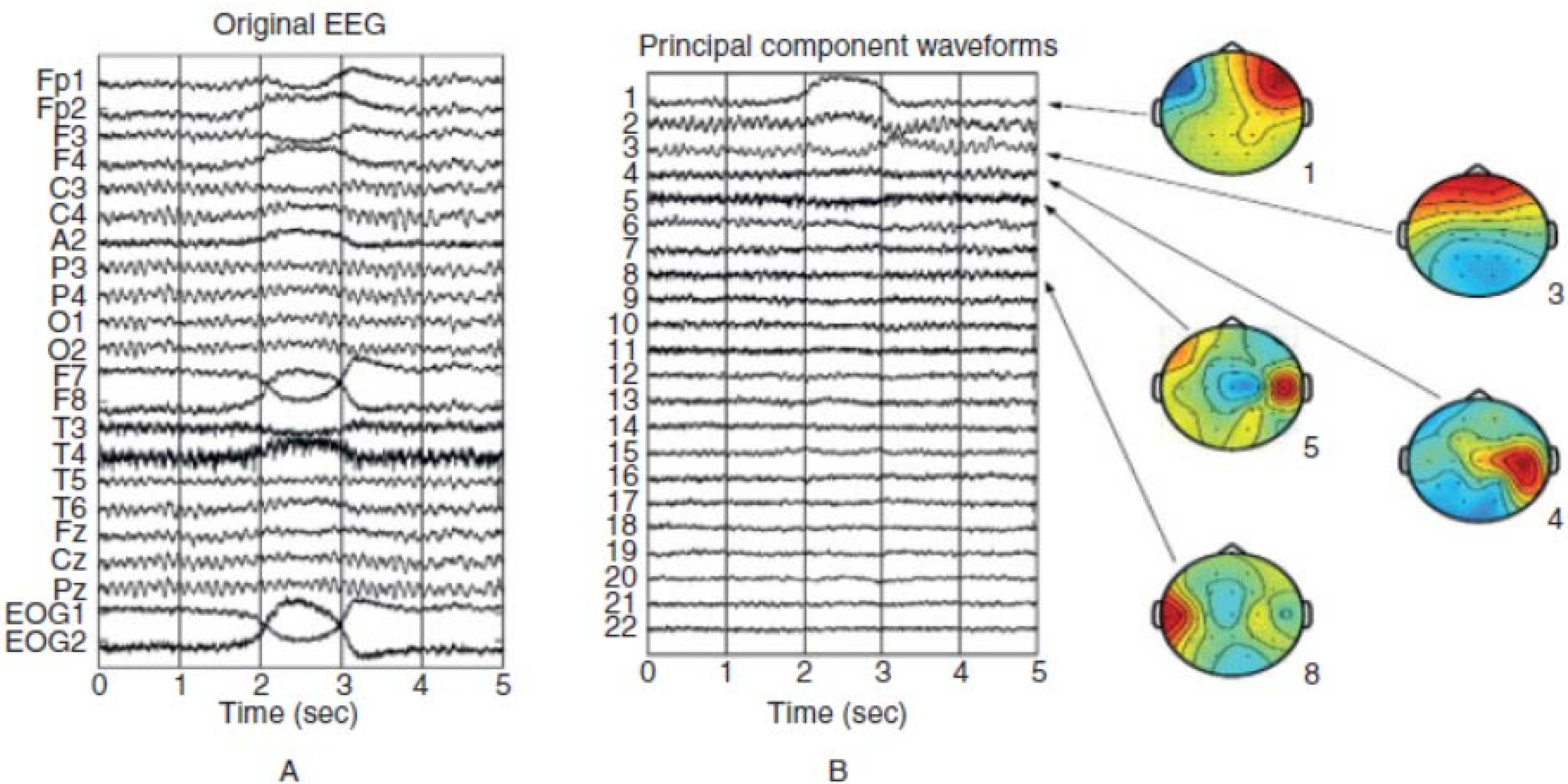


Figure 4.10. **PCA applied to EEG data.**

III.PRINCIPAL COMPONENT ANALYSIS

Principal component analysis (PCA) involves a mathematical procedure that transforms a number of (possibly) correlated variables into a (smaller) number of uncorrelated variables called principal components. The first principal component accounts for as much of the variability in the data as possible, and each succeeding component accounts for as much of the remaining variability as possible. Principal components are guaranteed to be independent only if the data set is jointly normally distributed. PCA is sensitive to the relative scaling of the original variables. Depending on the field of application, it is also named the discrete Karhunen–Loève transform (KLT), the Hotelling transform or proper orthogonal decomposition (POD).

P.A. Babu and K.V. S.V. R. Prasad, "Removal of Ocular Artifacts from EEG Signals Using Adaptive Threshold PCA and Wavelet Transforms," in *International Conference on Communication Systems and Network Technologies*, 2011, pp. 572-575.

PCA is the simplest of the true eigenvector-based multivariate analyses. Often, its operation can be thought of as revealing the internal structure of the data in a way which best explains the variance in the data. If a multivariate dataset is visualized as a set of coordinates in a high-dimensional data space (1 axis per variable), PCA can supply the user with a lower-dimensional picture, a "shadow" of this object when viewed from its (in some sense) most informative viewpoint. This is done by using only the first few principal components so that the dimensionality of the transformed data is reduced.PCA is sensitive to the scaling of the variables. If we have two variables and they have the same sample variables and are positively correlated, then the PCA will tend to rotate by 45^0 and the loadings for the two variables with respect to the principal components will be equal.PCA is mathematically defined as an orthogonal linear transformation that transforms the data to a new coordinate system such that the greatest variance by any projection of the data comes to lie on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on.

4.5 Spatial Filtering

4.5.3 Independent Component Analysis (ICA)

Lecturer: Minghui Shi, Xiamen University

4.5.3 Independent Component Analysis (ICA)

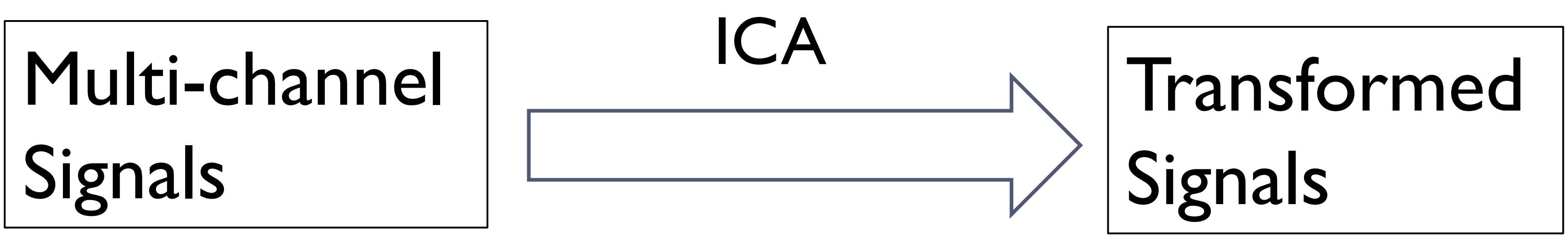
- (*read this section with the following questions.*)
- What's the limitation of PCA, and what's the goal of ICA?
- What do Mixing Matrix and Unmixing Matrix mean ?
- What are the two main techniques to compute unmixing matrix?
- What can we do by using ICA?

4.5.3 Independent Component Analysis (ICA)

- **What's the limitation of PCA?**
  - PCA produces a feature vector **a** that is both low-dimensional and decorrelated.
  - However, the resulting feature vector may still retain higher order statistical dependencies (beyond correlation).

4.5.3 Independent Component Analysis (ICA)

- **What's the goal of ICA?**
  - The goal of ICA is to find the independent components hidden in the raw signal.



4.5.3 Independent Component Analysis (ICA)

4.5.3 Independent Component Analysis (ICA)

x: observed signals **y:** Hidden independent sources

$$\mathbf{x} = M\mathbf{y}$$

mixing matrix

$$\boxed{\mathbf{y} = W\mathbf{x}}$$

unmixing matrix

4.5.3 Independent Component Analysis (ICA)

- ICA has proved useful in a variety of settings in BCI applications:
  - Use **y** as a feature vector in classification
  - Isolate interesting brain rhythms
  - Eliminate muscle artifacts in EEG

$$\mathbf{x} = \underbrace{M}_{\textit{mixing matrix}}\mathbf{y}$$

$$\mathbf{y} = \underbrace{W}_{\textit{unmixing}\ \text{matrix}}\mathbf{x}$$

4.5.3 Independent Component Analysis (ICA)

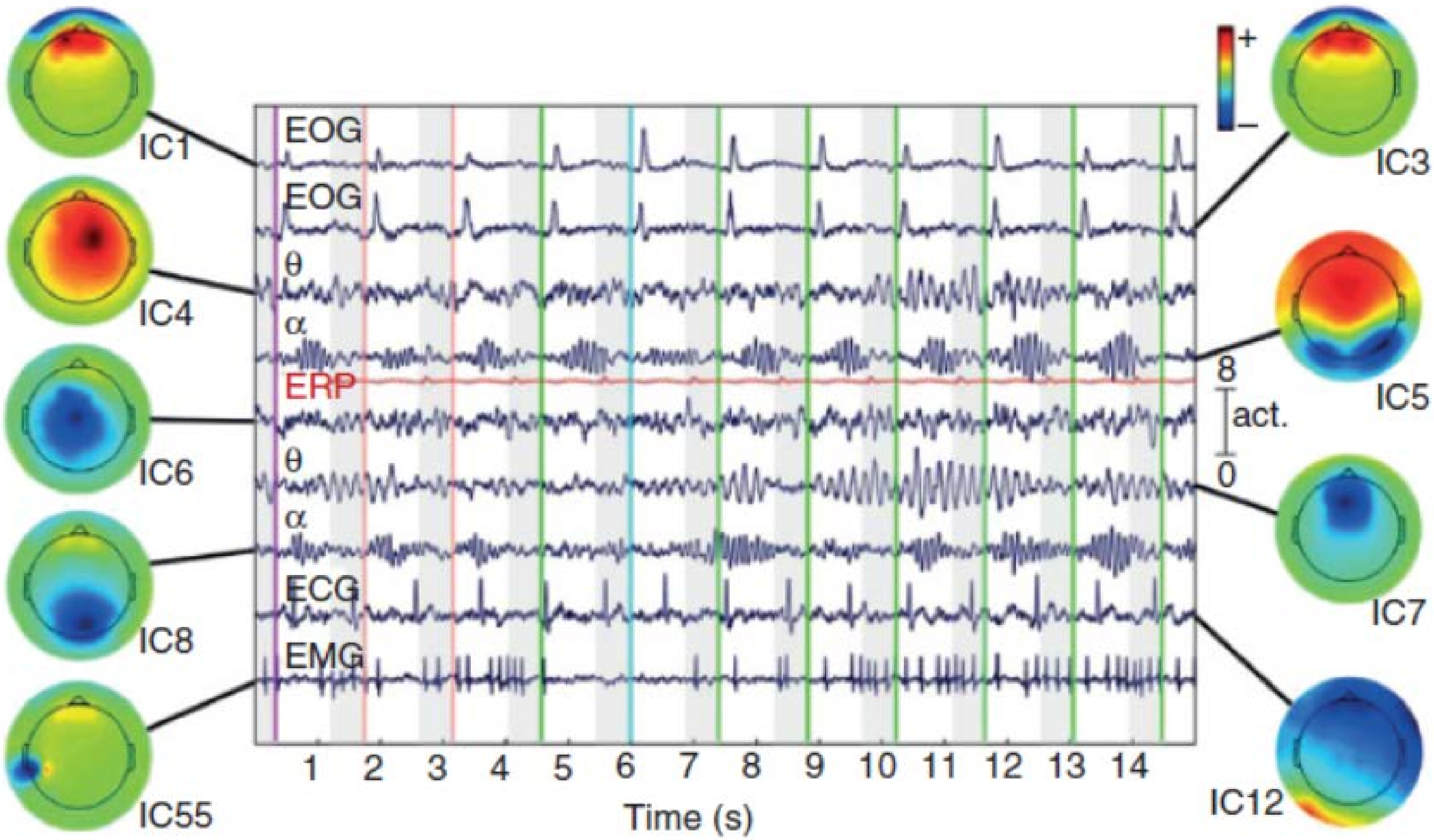
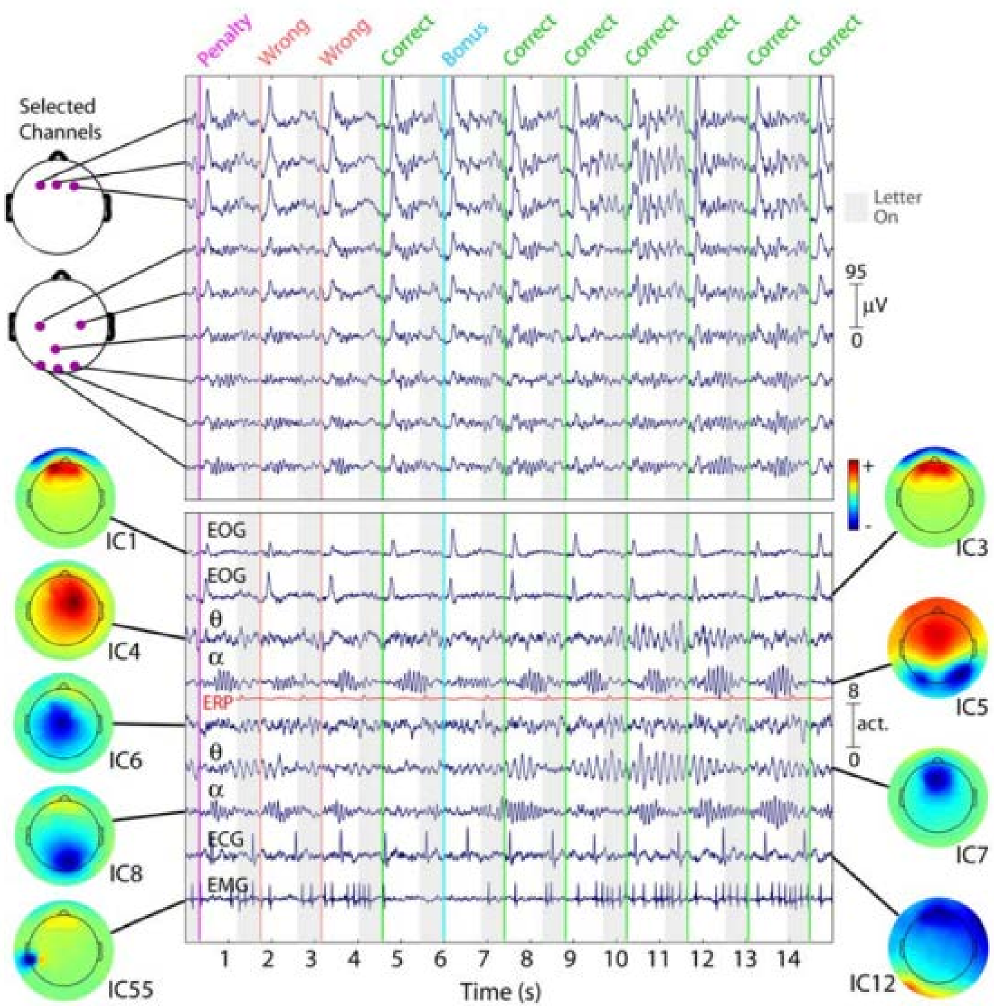


Figure 4.11. **ICA applied to EEG data.**

4.5.3 Independent Component Analysis (ICA)



J. Onton and S. Makeig, "Information-based modeling of event-related brain dynamics," *Progress in Brain Research*, vol. 159, pp. 99-120, 2006.

4.5 Spatial Filtering

4.5.4 Common Spatial Patterns (CSP)

Lecturer: Minghui Shi, Xiamen University

4.5.4 Common Spatial Patterns (CSP)

- (*read this section with the following questions.*)
- Compared to PCA and ICA,
  what's the significant difference between CSP and them?
- What's the principle of CSP?
- Why does CSP become a popular filtering method for EEG BCIs?
- What's the goal of CSP?
- How to reach the goal?

4.5.4 Common Spatial Patterns (CSP)

- **Compared to PCA and ICA, what's the significant difference between CSP and them?**
  - CSP is a supervised techniques.
  - The training dataset is labeled.
  - As an example, suppose we have collected brain data when the subject is performing two different tasks (e.g., hand versus foot motor imagery).

4.5.4 Common Spatial Patterns (CSP)

- **What's the principle of CSP?**
  - CSP finds spatial filters such that the variance of the filtered data from one class is maximized while the variance of the filtered data from the other class is minimized.
  - The resulting feature vectors thus enhance the discriminability between the two classes.

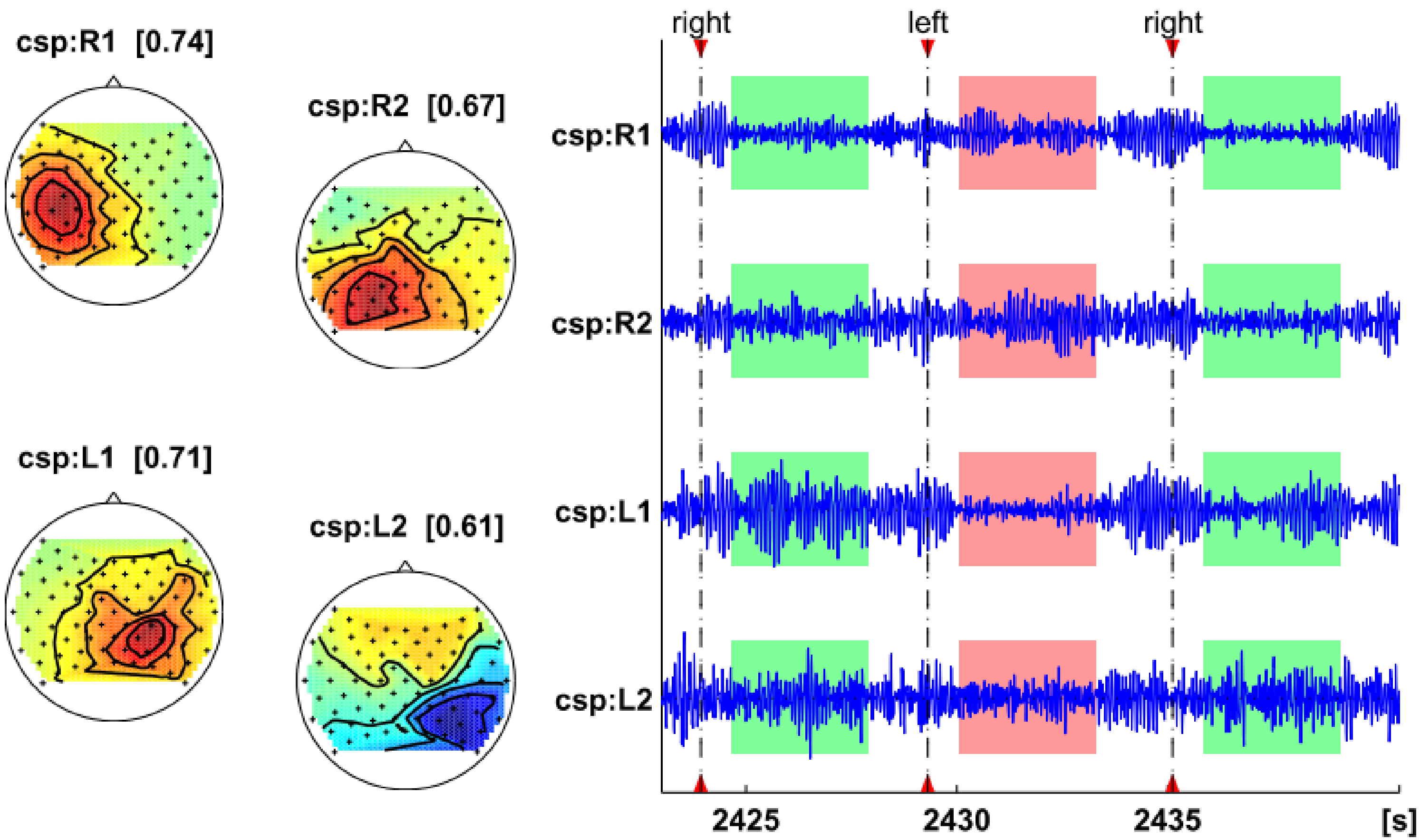
4.5.4 Common Spatial Patterns (CSP)

- **Why does CSP become a popular filtering method for EEG BCIs?**
  - CSP has emerged as a popular filtering method for EEG BCIs, especially those that rely on the power in a frequency band for control.
  - Since the variance of EEG signals filtered in a given frequency band corresponds to the power in this band,
  - and CSP finds spatial filters such that the variance of the filtered data from one class is maximized while the variance of the filtered data from the other class is minimized,
  - thus CSP essentially maximizes the discriminability of the features used in these BCI.

4.5.4 Common Spatial Patterns (CSP)

- **Video**
  - **Play a game with brain**

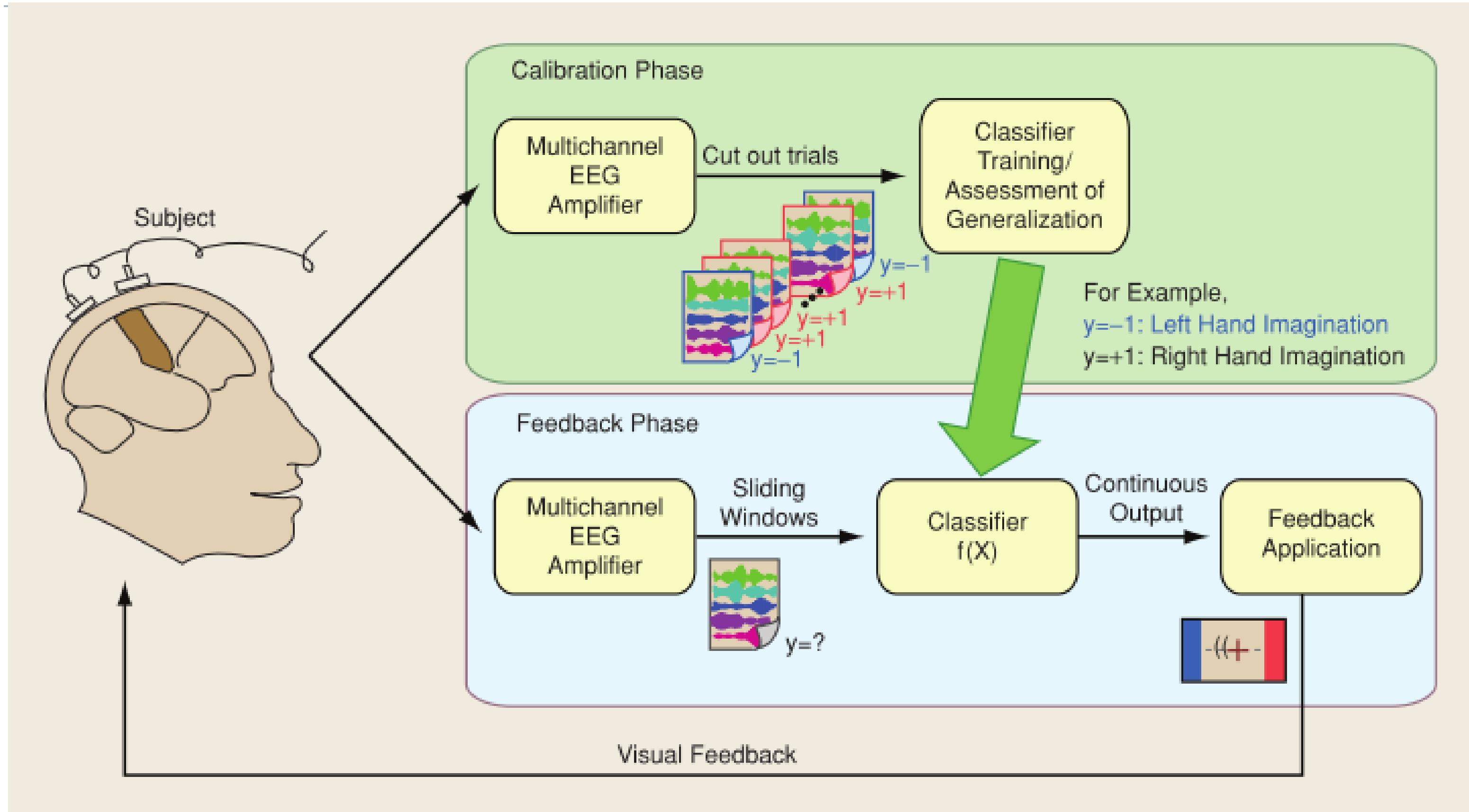
4.5.4 Common Spatial Patterns (CSP)



K. R. Müller, M. Tangermann, G. Dornhege, M. Krauledat, G. Curio, and B. Blankertz, "Machine learning for real-time single-trial EEG-analysis: From brain–computer interfacing to mental state monitoring," *Journal of Neuroscience Methods*, vol. 167, pp. 82-90, 2008.

Figure 4.12. **CSP applied to EEG data.**

4.5.4 Common Spatial Patterns (CSP)



Overview of the machine-learning-based BCI system

B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Muller, "Optimizing spatial filters for robust EEG single-trial analysis," Signal Processing Magazine, IEEE, vol. 25, pp. 41-56, 2008.

Chapter 4
Signal Processing

4.6 Artifact Reduction Techniques

Lecturer: Minghui Shi, Xiamen University

4.6.0 Artifact Reduction Techniques

- (*read this section with the following questions.*)
- What are artifacts in BCIs?
- Can artifacts be useful?
- Why do we reduce artifacts?
- In what two ways can we reduce artifacts?
- What kinds of artifacts are originated from within the subject's body?

4.6.0 Artifact Reduction Techniques

- What are artifacts in BCIs?
  - Artifacts in BCIs are any undesirable signals that originate from outside the brain.
  - For example, in EEG BCIs, one often encounters 50/60Hz power-line noise and artifacts caused by muscle or eye movements.

- Can artifacts be useful?
  - Some of these artifacts may be permissible or even exploited as control signals for certain applications such as gaming or novel modes of human-computer interaction.

4.6.0 Artifact Reduction Techniques

- **Why do we reduce artifacts?**
  - A true brain-computer interface should possess the ability to eliminate or reduce such artifacts so that the signals being used to control a device originate solely from the brain.
  - Signal-processing techniques can be used to achieve this goal.

- **In what two ways can we reduce artifacts?**
  - Hardware: Faraday cage
  - Software: filtering techniques

4.6.0 Artifact Reduction Techniques

- What kinds of artifacts are originated from within the subject's body?
  1. rhythmic artifacts due to respiration and heartbeat(ECG).
  2. signal distortion or attenuation due to skin conductance changes.
  3. eye movement and eye blink artifacts (electro-oculographic or EOG), which appear as high-amplitude deflections in brain signals such as EEG with frequencies in the range 3–4Hz.
  4. muscle artifacts (electromyographic or EMG) caused by movements of the head, face, jaw, tongue, neck, and other parts of the body; EMG artifacts tend to occur maximally in the 30Hz or higher frequency range.

4.6 Artifact Reduction Techniques

4.6.1 Thresholding

4.6.1 Thresholding

- (*read this section with the following questions.*)
- What's the thresholding method to reduce artifacts?
- How to determine the threshold?
- What's the drawback of the thresholding method?
- What's the complementary approach?

4.6.1 Thresholding

- **What's the thresholding method to reduce artifacts?**
  - If the magnitude or some other characteristic of a recorded EOG or EMG signal exceeds a pre-determined threshold, the brain signals recorded during that epoch are deemed to be contaminated and rejected.
- **How to determine the threshold?**
  - e.g., asking the subject to make various kinds of eye or body movements to calibrate the threshold.
- **What's the drawback of the thresholding method?**
  - not all artifact-contaminated data may be rejected by this method.

4.6.1 Thresholding

- **What's the complementary approach?**
  - **Artifact removal methods**：
    - Identify and excise artifacts from data while preserving neurological phenomena useful for BCI.

4.6 Artifact Reduction Techniques

4.6.2 Band-Stop and Notch Filtering

Lecturer: Minghui Shi, Xiamen University

4.6.2 Band-Stop and Notch Filtering

- (*read this section with the following questions.*)
- What's the principle?
- How to do it?
- What are the commonly used method?
- What's the limitation?

4.6.2 Band-Stop and Notch Filtering

- **What's the principle?**

1. **attenuate** the components of a signal in a specific frequency band;
2. **pass** the rest of the components of the signal.

- **How to do it?**

1. **transform** the signal to the frequency domain (e.g., using FFT);
2. **filter out** the desired frequency band;
3. **transform** back to the time domain.

4.6.2 Band-Stop and Notch Filtering

- **What are the commonly used method?**
  1. Notch filter is set to 59-61 Hz → Power line (in the U.S.)
  2. Notch filter is set to 1-4 Hz →EOG
  3. Low-pass filtering →EMG

- **What's the limitation?**
  - Filtering approaches work only when the brain signal of interest does not fall within the frequency range of artifacts.
  - For example, low-pass filtering may remove EMG artifacts, but if the BCI utilizes high-frequency components of the brain signal, such filtering may eliminate these useful components as well.

4.6 Artifact Reduction Techniques

4.6.3 Linear Modeling

4.6.3 Linear Modeling

- (*read this section with the following questions.*)
- What's the assumption?
- How does it remove EOG?
- Why is it not effective for removing EMG?

4.6.3 Linear Modeling

- **What's the assumption?**
  - Assume that the effect artifacts is additive.

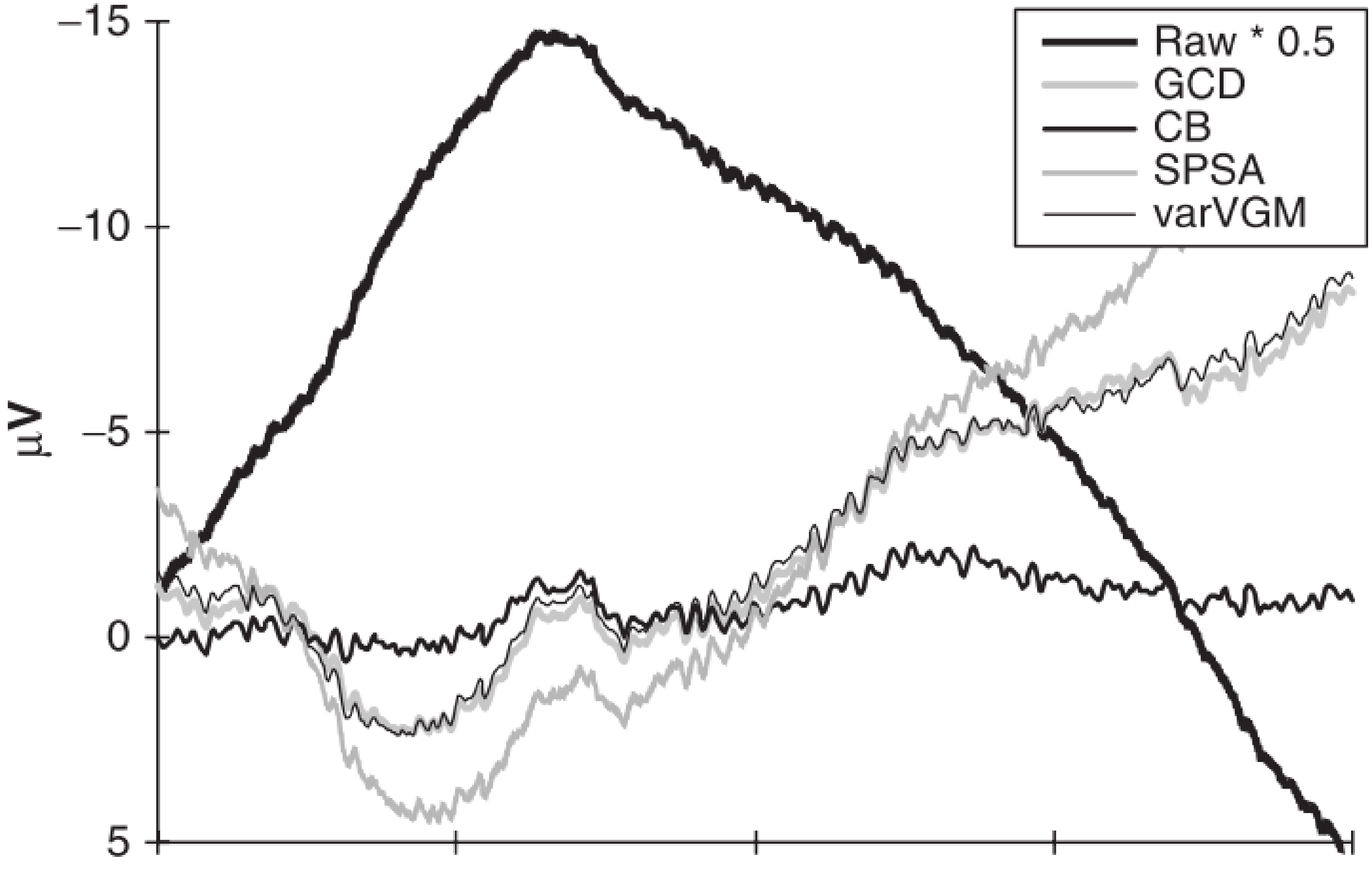
4.6.3 Linear Modeling

How does it remove EOG?

$$EEG_i(t) = EEG_i^{true}(t) + K \cdot EOG(t)$$

$$EEG_i^{true}(t) = EEG_i(t) - K \cdot EOG(t)$$

$$EEG_i(t) = EEG_i^{true}(t) + K \cdot EOG(t) \qquad EEG_i^{true}(t) = EEG_i(t) - K \cdot EOG(t)$$



R. Croft and R. Barry, "Removal of ocular artifact from the EEG: a review," *Neurophysiologie Clinique/Clinical Neurophysiology*, vol. 30, pp. 5-19, 2000.

R. J. Croft, J. S. Chandler, R. J. Barry, N. R. Cooper, and A. R. Clarke, "EOG correction: A comparison of four methods," *Psychophysiology*, vol. 42, pp. 16-24, 2010.

4.6.3 Linear Modeling

- **Why is it not effective for removing EMG?**
  - EMG artifacts arise from multiple muscle groups.

4.6 Artifact Reduction Techniques

4.6.4 Principal Component Analysis (PCA)

Lecturer: Minghui Shi, Xiamen University

4.6.4 Principal Component Analysis (PCA)

- PCA can find the directions of maximum variance in the recorded brain data (as discussed in Section 4.5.2), and is always combined with other methods to remove artifacts.

- References:

O. G. Lins, *Ocular artifacts in recording EEGs and event related potentials*, 1993.

P.A. Babu and K.V. S.V. R. Prasad, "Removal of Ocular Artifacts from EEG Signals Using Adaptive Threshold PCA and Wavelet Transforms," in *International Conference on Communication Systems and Network Technologies*, 2011, pp. 572-575.

T. Jung, S. Makeig, C. Humphries, T. Lee, M. J. Mckeown, V. Iragui, *et al.*, "Removing electroencephalographic artifacts by blind source separation," *Psychophysiology*, vol. 37, pp. 163-178, 2010.

4.6 Artifact Reduction Techniques

4.6.5 Independent Component Analysis (ICA)

Lecturer: Minghui Shi, Xiamen University

4.6.5 Independent Component Analysis (ICA)

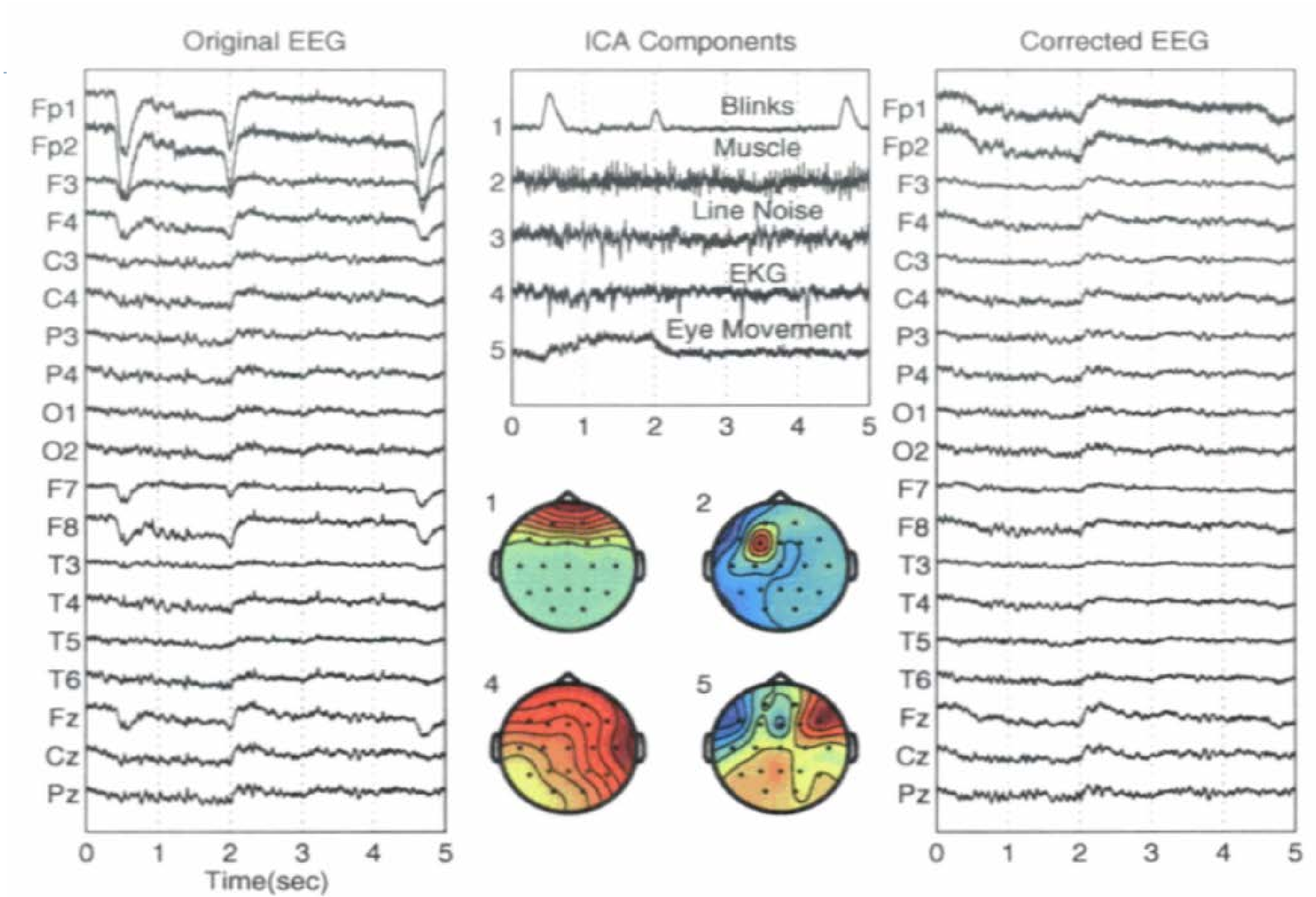
- Principle:

1. **Decomposes** brain signals (e.g., EEG) from a set of electrodes into a set of "components", using ICA.

2. **Identify** components due to EOG, EMG, or other artifacts;

3. **Reconstitute** the brain signal without these components.

4.6.5 Independent Component Analysis (ICA)



T. P. Jung, C. Humphries, T. W. Lee, S. Makeig, M. J. Mckeown, V. Iragui, *et al.*, "Extended ICA Removes Artifacts from Electroencephalographic Recordings," *Advances in Neural Information Processing Systems*, vol. 10, pp. 894-900, 1998.

4.6.5 Independent Component Analysis (ICA)

- Other references:

  - T. Jung, S. Makeig, C. Humphries, T. Lee, M. J. Mckeown, V. Iragui, *et al.*, "**Removing electroencephalographic artifacts by blind source separation**," *Psychophysiology*, vol. 37, pp. 163-178, 2010.

  - S. Makeig, S. Enghoff, T. P. Jung, and T. J. Sejnowski, "**Moving-window ICA decomposition of eeg data reveals event-related changes in oscillatory brain activity**," pp. 627--632, 2000.

Chapter 4
Signal Processing

4.7 Summary

4.7 Summary

- We reviewed techniques to
- Extract useful signals
- 4.1 Spike Sorting 39
- 4.2 Frequency Domain Analysis 40
  - 4.2.1 Fourier Analysis 40
  - 4.2.2 Discrete Fourier Transform (DFT) 43
  - 4.2.3 Fast Fourier Transform (FFT) 45
  - 4.2.4 Spectral Features 45
- 4.3 Wavelet Analysis 45
- 4.4 Time Domain Analysis 46
  - 4.4.1 Hjorth Parameters 46
  - 4.4.2 Fractal Dimension 48
  - 4.4.3 Autoregressive (AR) Modeling 49
  - 4.4.4 Bayesian Filtering 49
  - 4.4.5 Kalman Filtering 52
  - 4.4.6 Particle Filtering 54
- 4.5 Spatial Filtering 54
  - 4.5.1 Bipolar, Laplacian, and Common Average Referencing 55
  - 4.5.2 Principal Component Analysis (PCA) 56
  - 4.5.3 Independent Component Analysis (ICA) 60
  - 4.5.4 Common Spatial Patterns (CSP) 61

4.7 Summary

4.7 Summary (Con.)

4.7 Summary (Con.)

Relative Video

- **Machine Learning for Brain-Computer Interfaces**
- http://videolectures.net/nips09_hill_mlb/?q=Brain%20computer%20interface

Thank you for your attention!